GHOST DATA STILL EXISTS

The student-friendly challenge guide

In simple words: build an AI agent that checks whether a person's data was really deleted after they asked a company to delete their account.

Imagine this: You delete your account from an online shopping app. The app says, “Your account has been deleted.” A week later, you try to sign up again, but it says your email is already registered. How? Your data may still be hiding in other parts of the company's systems.

That is exactly the problem you need to solve.

1. What is the challenge asking you to build?

Build an **AI-powered deletion checker**. It should investigate the evidence provided by the company and decide whether a user's personal data has actually been removed.

Your agent must:

- Find the user's current and previous identifiers (such as old email addresses).
- Search the supplied evidence to find copies of their data.
- Detect systems or automatic processes that could bring the data back.
- Decide what should be deleted, hidden, anonymized, or kept according to the rules.
- Give the company an ordered action plan and checks to prove the result.

Important: A dashboard saying “Complete” does NOT prove that deletion is complete. Your agent must check the evidence itself.

2. The story behind the problem

The imaginary company **Northstar Shop** loses $2.4 million after a customer who was supposedly deleted cannot create a new account because their email is still registered. The investigation discovers that personal data may remain in secondary databases, support files, search indexes, analytics exports, logs, archives, and backups. A sync process may even recreate data after it was deleted from the main account system.

Your job is to look beyond the main account database. Check all relevant evidence and work out the true situation at the specified audit time.

3. What files are provided?

The repository contains a **dataset/** folder. These files are your investigation material:

FILE	WHAT IT CONTAINS
deletion_requests.csv	The cases you must investigate.
sample_deletion_requests.csv	A small set of example cases to test your solution.
sample_output.csv	Example of correctly formatted answers. Use it as a format guide.
user_identities.csv	Users' current identity details.
user_audit_log.csv	Past changes to user identity details.
system_activity.csv	Activity such as deletion, syncing, exporting, backups, restoring, and indexing.
deletion_rules.csv	Rules explaining what must be deleted, redacted, anonymized, or retained.
evidence_files.csv	A list of the evidence files available.
output.csv	The blank file where your final answers go.

Evidence can also be stored as PDFs, images, logs, JSON files, or SQL database snapshots inside **dataset/evidence/**. Some snapshots may be from secondary or backup systems, not just the main database.

4. How do you decide if deletion is complete?

A case is marked **COMPLETE** only when all four conditions are true:

1. Every relevant copy has received the treatment required by the deletion rules.
2. No active sync, import, or restore process can recreate the user's data.
3. Any records that are allowed to remain contain only the identifiers permitted by the rules.
4. Records that the company is required to keep are still available.

The three possible answers

STATUS	MEANING	WHAT TO INCLUDE
COMPLETE	The evidence proves deletion is finished.	Only verification checks that prove the current state.
INCOMPLETE	Some data remains, or a risk still exists.	What you found, the full fix plan, and checks to run after fixing it.
NOT_APPLICABLE	Deletion was not requested or is not authorized for this case.	An empty list: **[]**

5. What should your AI agent look for?

Think of the investigation in four parts:

CHECK	IN PLAIN ENGLISH
Remaining records	Is the user's data still stored somewhere it should have been removed from?
Exposed personal information	Does a record that is allowed to remain still reveal details like name, email, or phone number?
Active data flows	Is a sync, import, or other process still copying or recreating the data?
Restore risks	Could a backup or restore operation bring deleted data back?

Special case: account tombstones

Some systems keep a small account reference (called a **tombstone**) so records the company is legally or operationally required to retain can still refer to the same internal user. The internal user ID may remain, but direct identifiers such as name, email, phone number, and email hash must be removed when the rules require it.

6. What actions can your solution recommend?

The challenge requires exact operation names in the final output. The table below explains them in student-friendly language. **Use the exact capitalized labels** in your code and output.

PHASE	EXACT OPERATION	SIMPLE MEANING
FINDING	REPORT_REMAINING_RECORD	Report a record that should have been deleted but still exists.
FINDING	REPORT_EXPOSED_PII	Report personal details that should have been removed from a retained record.
FINDING	REPORT_ACTIVE_DATA_FLOW	Report a process that can copy or recreate the data.
FINDING	REPORT_RESTORE_RISK	Report a backup or restore path that could bring the data back.
REMEDIATION	DISABLE_DATA_FLOW	Stop the process that copies or recreates data.
REMEDIATION	BLOCK_RESTORE	Prevent the affected backup from restoring the data.
REMEDIATION	DELETE_RECORD	Remove the entire record.
REMEDIATION	REDACT_RECORD	Keep the record but remove identifying details.
REMEDIATION	ANONYMIZE_RECORD	Keep the record but permanently make it impossible to link to the user.
VERIFICATION	VERIFY_RECORD_ABSENT	Check that the record is gone.
VERIFICATION	VERIFY_RECORD_PRESENT	Check that a record the company must keep is still there.
VERIFICATION	VERIFY_RECORD_REDACTED	Check that required identifying details have been removed.
VERIFICATION	VERIFY_RECORD_ANONYMIZED	Check that the record can no longer be linked to the user.
VERIFICATION	VERIFY_DATA_FLOW_DISABLED	Check that the copying process is inactive.
VERIFICATION	VERIFY_RESTORE_BLOCKED	Check that the data cannot be restored.

Do not invent new operation names or mix up the operation and state values. The evaluator expects the exact combinations specified in the official brief.

7. What should the final output look like?

Your solution must create a file called **output.csv** with exactly these three columns, in this order:

```
case_id,deletion_status,steps_json
```

- **case_id:** The ID of the case you investigated.
- **deletion_status:** COMPLETE, INCOMPLETE, or NOT_APPLICABLE.
- **steps_json:** A JSON list containing your findings, actions, and verification checks in the required order.

Each step contains details such as the step number, phase, system, resource, operation, state, and evidence ID. Follow the exact structure shown in the original participant brief and sample_output.csv.

Important output rules

- There must be exactly **518 data rows**, plus the header row.
- Include one row for every request in deletion_requests.csv.
- Sort all rows by case_id.
- Step numbers start at 1 and increase without gaps.
- For COMPLETE cases, include verification steps only, supported by evidence.

- For INCOMPLETE cases, include findings, the full remediation plan, and future verification steps.
- For NOT_APPLICABLE cases, steps_json must be **[]**.

8. How should you approach the challenge?

A practical workflow for your team

STEP 1 — Understand the data

Read the repository instructions and inspect the CSV files. Start with the sample requests and sample output.

STEP 2 — Identify the person

Match the user's current and historical identifiers before searching for their records.

STEP 3 — Investigate the evidence

Check the primary system, secondary systems, replicas, logs, exports, indexes, archives, and backups that are relevant to the case.

STEP 4 — Check the rules

Use deletion_rules.csv to decide whether each record should be deleted, redacted, anonymized, or retained.

STEP 5 — Find risks

Look for active sync/import processes and backup or restore paths that could bring data back.

STEP 6 — Decide the status

Mark the case COMPLETE, INCOMPLETE, or NOT_APPLICABLE based on the evidence and rules.

STEP 7 — Build output.csv

Create one correctly formatted row for each of the 518 cases, with correctly ordered steps.

STEP 8 — Test and package

Validate your CSV, evaluate your solution, write the required reports, and prepare code.zip.

9. What do you need to submit?

SUBMISSION	WHAT IT SHOULD CONTAIN
code.zip	Your runnable solution, prompts, configuration, README, and evaluation/ folder. Do not include the dataset or nested archives.
output.csv	Exactly 518 answer rows, sorted by case_id.
chat_transcript	The development conversation log produced by your AI coding tool. Follow the repository's AGENTS.md instructions.
AI interview	Explain your approach, system design, and how you evaluated your solution.

Two reports must be inside code.zip

evaluation/evaluation_report.md — Include case-level accuracy, step-level and field-level accuracy, sequence accuracy, at least three extra test cases created by your team, and the failures/improvements you found.

evaluation/usage_report.md — Explain the final full-dataset run: model providers and names, number of model calls, input/output/total tokens, average tokens per request, and estimated total and per-request cost. Include per-model totals if you used multiple models.

10. Final checklist before submission

■ output.csv has the exact header: case_id,deletion_status,steps_json.
■ There are exactly 518 data rows, sorted by case_id.

- ■ Every steps_json value is valid JSON and step numbers are consecutive.
- ■ Every operation/state combination is allowed and steps are correctly ordered.
- ■ COMPLETE cases contain evidence-backed verification steps only.
- ■ INCOMPLETE cases contain findings, a complete fix plan, and future checks.
- ■ NOT_APPLICABLE cases contain [].
- ■ code.zip contains the runnable code, README, and both evaluation reports.
- ■ The chat transcript follows the required format.
- ■ No dataset, virtual environment, cache, API keys, passwords, or other secrets are included.

The simplest way to remember the challenge

Find the data → Check the rules → Stop it from coming back → Prove the result.

This guide simplifies the six-page participant brief. For exact field names, operation/state combinations, evidence rules, ordering, and any discrepancy, the official repository and full problem statement take priority.

Official repository: github.com/interviewstreet/hackerrank-orchestrate-october26
Challenge: hackerrank.com/contests/hackerrank-orchestrate-october26/challenges/ghost-data-still-exists